Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/08/2025
PI Name:	Liao, Ya-Ting Ph.D.		
Project Title:	Predicting Material Flammability in Partial Gravity Using Microgravity and Ground Data		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-PSI E.8 Physical Sciences Informatics
Start Date:	11/01/2023	**End Date:**	11/02/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
Contact Monitor:	Johnston, Michael	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/2/2026 per F. Hernandez/NASA (Ed., 4/8/26). NOTE: Start and end dates have changed per NSSC grant documents. The new period of performance is 11/1/2023 - 11/2/2025 (Ed., 5/20/24).		
Key Personnel Changes/Previous PI:	Dr. Ankit Sharma, the former Postdoctoral researcher in the Principal Investigator's lab at Case Western Reserve University (CWRU), left CWRU in January 2024. A new postdoctoral researcher joined the PI's team in October 2024. She replaced Dr. Sharma on this project.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC24K0310		
Performance Goal No.:			
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Task Description:	As NASA and commercial space companies expand their exploration into deeper space, spacecraft fire safety has become more important than ever. While knowledge of fire dynamics in microgravity has accumulated over the past few decades, there is a knowledge gap about how fires behave and how materials burn in partial gravity environments (e.g., on the Moon or Mars surfaces). Previous research has shown that material flammability and fire risk can be greater in partial gravity than both normal and microgravity. To ensure safety and success of future space missions, there is an urgent need to advance the understanding of fire dynamics in partial gravity. In this project, it is hypothesized that burning behaviors and flammability boundaries of solid materials in partial gravity can be predicted using data collected in normal and microgravity. The overall goal of this project is to leverage the rich dataset collected in previous microgravity experiments to advance understanding of fire behaviors in partial gravity. The microgravity data will be complemented by existing and new ground data to lead to new science. Ultimately, we want to answer the following questions. “What can we learn from the microgravity data obtained in different experiments and different ambient conditions regarding material flammability in partial gravities?” “How do we predict burning behaviors in partial gravity using data collected in normal and microgravity?” The project consists of three major components. The first component is analysis of previous microgravity data. Tools will be developed to analyze data available on the NASA’s Physical Sciences Informatics (PSI) system for microgravity combustion experiments (Burning and Suppression of Solids/BASS, BASS-II; Spacecraft Fire Safety Experiments: SAFFIRE-I, SAFFIRE-II, and SAFFIRE-III). Flame characteristics including spread rate, length, appearance, color, and flammability boundaries will be extracted, documented, and compared. Other microgravity datasets (e.g., International Space Station/ISS Confined Combustion project) will also be considered. This will result in a comprehensive inter-project meta-analysis of the current knowledge on microgravity combustion. The second component of the project is ground experiments. Burning experiments will be performed in a combustion chamber in the Principal Investigator (PI) lab with different environmental conditions. The experimental setup and sample selections will resemble the PSI microgravity experiments, enabling direct comparisons of ground and space data. This will allow us to correlate the gravity-induced buoyancy flows (in normal gravity) to the imposed flows in microgravity, in terms of their effects on the burning process. The third component will be numerical simulations. Both microgravity and ground experiments will be simulated. After the model is validated against the experimental data, the detailed profiles obtained in the numerical simulations will be used to understand underlying physics of the experimental observations. The model will then be able to simulate solid burning in different partial gravities, leading to a more complete understanding of the role of buoyancy flow on fire behavior. Through these efforts, we will identify invariant dimensionless parameters for flame spread and develop correlations between burning characteristics and the identified dimensionless parameters. These dimensionless parameters will incorporate various variables, including sample dimensions (e.g., width, thickness); gravity; and ambient conditions (pressure, flow speed, oxygen percentage). All generated outcomes will be made available for future investigators to test theories or train models.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The expected outcomes of the proposed project include 1) a comprehensive summary/comparison of flame characteristics (e.g., flame length, flame color and appearance, spread rate, flammability limits) across various microgravity investigations (e.g., BASS, SAFFIRE, Confined Combustion), 2) newly generated ground test results that can be directly compared to existing microgravity data, 3) numerically simulated detailed flame profiles in different pressures, gravities, flow speeds, and oxygen percentages, 4) identification of invariant dimensionless parameters for flame spread, and 5) correlations between burning characteristics and the identified dimensionless parameters which incorporating sample dimensions (e.g., width, thickness), gravity, and ambient conditions (pressure, flow speed, oxygen percentage). All generated outcomes will be made available for future investigators to test theories or train models. This directly contributes to the advancement of fire science. The developed understanding of burning behaviors in partial gravity will help improve the test methods and design of future space vehicles. The ultimate goal of this project is to reduce tragic loss of life and property both on earth and in space.
	The effort of this project focuses on the analysis of previous microgravity data on NASA’s Physical Sciences Informatics (PSI) system. Leveraging the synergy of the Principal Investigator (PI)’s two other concurrent projects (from NASA and National Science Foundation/NSF), this project also involves the ground experiments in the PI’s lab and numerical modeling. 1) Analysis of the Micro- and Partial Gravity Data The major component of this project is the analysis of data available on the PSI system for microgravity combustion experiments. Undergraduate student, Nathan Kralik, was recruited to support this project. He joined the PI’s lab in June 2024 as a summer student researcher. Mr. Kralik continued working on the project and became a BS/MS student in Fall 2024. In this reporting period, we continued the efforts on literature review, categorization of the PSI data, and development of a MATLAB code to process the PSI data. We have focused on the BASS and BASS II experiments. In these experiments, various materials (e.g., (e.g., polymethyl methacrylate/PMMA, and a cotton-based fabric called SIBAL) were burned under controlled, concurrent air flow in microgravity, and images and videos were taken during the burning process. We successfully developed a MATLAB code to track the flame progression from the images and videos in the database. This was accomplished by creating an edge tracing program to trace the flame in each image, and then a location tracking program to track the movement of the trace over time. Note that the detected flame edge and tracked locations of the flame depend on a threshold value set in the code. An optimal value was determined for each experiment for accurately detecting the flame edge, while minimizing interference from background noise. Additionally, a subroutine was written to process the texts overlaid on each flame image and to obtain information on the real-time air flow speed in the experiments. The extracted air flow speed, and the tracked flame locations versus time, were obtained for most of the tests in BASS and BASS II. Collaborating with Dr. Paul Ferkul from NASA Glenn Research Center, we applied our flame analysis program on NASA’s recent partial-gravity fire experiment, Lunar-g Combustion Investigation (LUCI). In LUCI, PMMA and SIBAL fabric were burned in simulated lunar gravity in a spinning sounding rocket. By including results from PSI,

Task Progress:

LUCI, and previous experiments in literature, flame characteristics will be mapped against different gravity levels and environmental conditions. An abstract was submitted and accepted for presentation in the 2025 Meeting of the American Society for Gravitational and Space Research (ASGSR) on this effort.

2) Ground Experiments

Burning experiments were performed in a combustion chamber in the PI's lab with different environmental conditions. A Ph.D. student (Robin Neupane) updated and restored the functionality of an old combustion chamber on loan from NASA Glenn Research Center. We conducted burning experiments in reduced pressures and in different oxygen percentages. The goal was to artificially lower the buoyant flows (which occurs in a partial gravity environment) via reducing the ambient gas density and to explore the burning characteristics of solid materials near the extinction limits.

In the first reporting period, upward flame spread experiments were performed at different pressures (ranging from 10 kPa to 100 kPa) and oxygen concentration environment (9% - 21%) for thin cellulose-based solid samples (filter paper). As pressure increased from the extinction limit, different burning behaviors were observed: no ignition, partial flame spread, steady flame spread, and accelerating flame spread. A similar trend was observed as the ambient oxygen concentration increased. In partial pressure conditions (e.g., 25-50 kPa), flames exhibited characteristics that are typically observed in micro- and partial gravity environments: blue and dim. Flame spread rate and sample burnt length were deduced and compared between different pressure and oxygen levels. Overall, the burning intensity and the flame spread rate decreased with the decrease in ambient pressure and oxygen. The decrease in flame spread rate at reduced pressure was attributed to increase in flame standoff distance and decrease in convective heat transfer to the solid, whereas the decrease in flame spread rate in reduced oxygen concentration environment was attributed to decrease in flame temperature.

Current and previous studies from literature performed at different ambient environments were correlated using the concept of flame standoff distance (δf), which was estimated using the theoretical viscous boundary layer thickness (δv). It was found that approximating $\delta f \sim \delta v$ for forced flow and $\delta f \sim 1/3\ \delta v$ for natural flow can predict the flame spread rate reasonably well for data obtained in micro-, partial, and normal gravities, for a wide range of environmental conditions away from extinction limits.

In the second reporting period, three undergraduate students (under the supervision of the Ph.D. student) conducted experiments of upward flame spread over thin PMMA (0.2 mm in thickness) in different ambient pressures (ranging from 40 kPa to 100 kPa). Custom MATLAB code, similar to that developed for the PSI data, was used to analyze the recorded flame spread videos and to deduce various fire characteristics (e.g., flame spread rate). The results were documented in the students' senior capstone project report and presented in the 2025 Spring Intersections Research Showcase event at Case Western Reserve University.

We also conducted downward flame spread experiments for thin cellulose-based solid fuel to complement the upward flame spread data obtained in the previous reporting period. The results were analyzed and compared to previous upward/downward flame spread data for same and similar materials (both from our own experiments and from literature). We are currently conducting theoretical analysis to correlate the downward flame spread data from the current and previous studies. Flame spread rates will be eventually predicted using a proposed correlation for buoyant-flow, purely-forced flow, and mixed flow conditions.

In this reporting period, we presented results from our experimental effort in a journal paper (Combustion and Flame) and in two conferences (the US National Meeting of the Combustion Institute in March 2025 in Boston Massachusetts, and the 2024 International Mechanical Engineering Congress and Exposition in November 2024 in Portland, Oregon). [Ed. Note: See Cumulative Bibliography.]

3) Numerical Simulations

The third component of this project is numerical simulation. For this effort, we use an in-house Computational Fluid Dynamics (CFD) combustion model that has been validated against various previous microgravity and ground experiments. A Ph.D. student (Jiaxuan Han) conducted two-dimensional transient numerical simulations to understand concurrent-flow flame spread over a thin solid sample in NASA's Spacecraft Fire Experiment (SAFFIRE) project (data available in the PSI). The sample material, model geometry, and ambient conditions were chosen based on the SAFFIRE VI experiment. A 94 cm-long cotton-fiberglass fabric blend (SIBAL fabric), situated at the center of a flow duct was burned in a concurrent forced flow of 20 cm/s. The ambient conditions were at 30 % oxygen and 0.54 atm pressure. The model consisted of both transient gas and solid phases. In the gas phase, full Navier-Stokes Equations were solved and a one-step, second-order Arrhenius reaction was included. In the solid phase, a two-step solid decomposition was considered. Combustion product dissociation effect at elevated oxygen conditions was also considered. The simulation results demonstrated that the flame spread rate, the location of the pyrolysis region, and the solid surface temperature distribution agreed well with those observed in the SAFFIRE VI experiment. The simulation results were also compared with simulation results of SAFFIRE IV for the same sample material burning at a different condition (23% oxygen and 1 atm pressure) to investigate the effect of ambient conditions on the fire behavior in microgravity. The numerical results successfully captured the increased flame spread rate observed in SAFFIRE VI than in SAFFIRE VI, despite a lower partial pressure of oxygen. The solid and gas profiles obtained in the numerical simulations were examined in detail. They were used to help explain the effects of pressure, oxygen, and the partial pressure of oxygen on flame spread.

In addition to simulations for SAFFIRE, the former Co-Investigator, postdoctoral researcher Dr. Ankit Sharma, used the three-dimensional version of the model to simulate upward flame spread over a thin solid sample in normal gravity. We first focused on validating the model against literature data. In these experiments, machine papers were burned in upward configuration in different oxygen (20-40%) and ambient pressure at 200 kPa in normal gravity. We found that, without the consideration of combustion product dissociation, the model over-predicted the flame temperatures and the flame spread rates at elevated oxygen environments. After the combustion product dissociation effects were considered and accommodated, the model was able to predict the flame spread rate reasonably well. Dr. Sharma left the PI's lab in February 2024. A new postdoctoral researcher, Dr. Nicharee Thinnakornsutibutr, was hired in October 2024 to continue this effort. Dr. Thinnakornsutibutr performed numerical simulations for upward flame spread over a thin solid material in different pressure and gravity environments. In particular, she used the model to evaluate the validity of pressure modeling, an approach adopted by many previous studies. Based on pressure modeling, fire behavior in partial gravity can be correlated to that in a reduced pressure environment if p2g are kept constant. Further analysis of the simulation results is ongoing.

	In addition to her own research, Dr. Thinnakornsutibutr helped mentor a high school student (Arianna Su) to conduct research in the PI's lab to support this project. Arianna developed a two-dimensional numerical model using Ansys Fluent to study buoyant flow induced by a vertical hot plate. By varying gravity and ambient pressure independently in the model, we investigated how buoyant flow and boundary layer thickness vary. In this reporting period, we presented results from our numerical effort in a journal article (in Applications in Energy and Combustion Science, article in press), a conference paper, a conference poster (both in the US National Meeting of the Combustion Institute in March 2025 in Boston Massachusetts), and a conference oral presentation (2024 Annual Meeting of the American Society for Gravitational and Space Research in December 2024 in San Juan, Puerto Rico).
Bibliography Type:	Description: (Last Updated: 10/10/2025)
Articles in Peer-reviewed Journals	Neupane R, Liao Y-T. "Correlating concurrent-flow flame spread rates in different pressure and oxygen conditions: Ground experiments and comparisons with previous micro-, partial, and normal gravities experiments." Combustion and Flame. 2025 Feb 1;272:113880. https://doi.org/10.1016/j.combustflame.2024.113880 , Feb-2025
Conference Materials (Downloadable)	Han J, Liao Y-T "Numerical study of flame spread over thin solids in reduced pressure and enhanced oxygen environments under microgravity condition" 14th U.S. National Combustion Meeting, Boston, Massachusetts, March 16-19, 2025. , Mar-2025 110072551 Han and Liao 14th_USNCM final.pdf
Conference Materials (Downloadable)	Neupane R, Liao Y-T. "Experimental study of downward flame spread over a thin sample in reduced and enhanced pressure." 14th U.S. National Combustion Meeting, Boston, Massachusetts, March 16-19, 2025. , Mar-2025 14th_USNCM_robinneupane_opposed_flame spread NSF.pdf
Conference Materials (Downloadable)	Thinnakornsutibutr N, Liao Y-T. "Fire scale modeling and effects of buoyant flow on upward flame spread." 14th U.S. National Combustion Meeting, Boston, Massachusetts, March 16-19, 2025. , Mar-2025 0311_Poster_Nicharee.pdf
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Conference Materials (Downloadable)	Neupane R, Liao Y-T. "Understanding the role of buoyant flow for accurate and robust scale modeling of upward flame spread." 2024 International Mechanical Engineering Congress and Exposition, Portland, Oregon, November 17-21, 2024. , Nov-2024 2024.11.IMECE Rising Start Poster.pdf
Conference Materials (Downloadable)	Neupane R, Liao Y-T. "Fire behavior and material flammability in reduced pressure environments." 2024 Ohio Space Forum, Cleveland OH, April 29-30, 2024. , Apr-2024 Robin et al osf poster.pdf
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Conference Materials (Downloadable)	Han J, Liao Y-T. "Numerical investigation of flame spread over a thin solid material in microgravity in enhanced oxygen and reduced pressure condition." 2024 Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024 2024.12.06 ASGSR - Han et al.pptx